Truth Maintenance, Belief Revision, and Correction Failure

Why Correcting a Fact Does Not Correct the System Built From It

Knowledge Systems and Reality Drift - Note #7 v1

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The Basic Pattern

Knowledge systems do more than store isolated facts. They derive conclusions, connect entities, populate reports, trigger decisions, and supply other systems. When an underlying claim changes, every dependent representation may also need to change.

Truth maintenance systems were developed to track the assumptions supporting a conclusion and revise beliefs when those assumptions are withdrawn. Belief revision studies how a knowledge base can incorporate new information while restoring consistency.

The open problem is not merely detecting that one statement is false. It is finding and unwinding everything that became true inside the system because that statement was once accepted.

When Correction Stops at the Source

A source record may be corrected while cached copies remain unchanged. A retracted scientific finding may persist in reviews, guidelines, databases, and model training data. A merged identity may be separated while scores derived from the merge remain in place.

Each derivative can be locally valid according to the information available when it was created. Without dependency tracking, the system cannot identify which outputs need reconsideration. The correction enters the representation stack through a narrower channel than the original claim.

This asymmetry creates correction failure. Information propagates widely because reuse is cheap. Retraction propagates slowly because every dependent system must recognize the change, locate affected conclusions, and decide how to revise them.

How Correction Debt Accumulates

Stage 1 - Assertion
A claim enters the knowledge system with some evidence or authority.

Stage 2 - Derivation
Rules, models, analysts, and applications generate conclusions that depend on the claim.

Stage 3 - Distribution
Those conclusions are copied into reports, caches, indexes, exports, and external systems.

Stage 4 - Revision
The original claim is corrected, weakened, or withdrawn.

Stage 5 - Correction Debt
Dependent representations persist because their lineage is missing, their update channels are weak, or revision is operationally expensive.

Belief Revision and Constraint Collapse

Correction debt is the accumulated set of claims, inferences, and decisions that no longer retain adequate support but continue operating. It is not identical to misinformation. The system may contain the correct source record while still acting through stale derivatives.

Constraint Collapse occurs when external correction can no longer travel through the structures shaped by the original error. The system retains coherence because downstream components continue agreeing with one another. Their agreement reflects shared ancestry rather than independent confirmation.

A correction-capable architecture requires claim-level provenance, dependency graphs, version-aware inference, expiration, re-evaluation triggers, and visible uncertainty. Every consequential abstraction should retain a live path through which omitted or revised reality can return as correction.

The Representation Stack

Observation becomes a claim. The claim becomes an inference. The inference becomes a summary, policy, score, or model input. The source is then corrected, but the correction does not automatically climb the same path.

This creates an uneven topology: broad channels for publication and narrow channels for retraction. The higher an unsupported conclusion has traveled, the more institutional work is required to make the system forget it.

Examples Across Systems

Science: Retracted findings remain cited through papers that copied the claim without preserving its evidentiary dependency.

Healthcare: A corrected diagnosis may remain embedded in risk scores, problem lists, referrals, and insurance decisions.

Knowledge Graphs: A deleted edge may leave inferred relations or cached query results intact.

Artificial Intelligence: Corrected web sources may not alter model weights, vector indexes, generated summaries, or synthetic texts derived from the earlier claim.

Conclusion

Correction is not complete when the originating fact changes. It is complete only when the system can identify and reconsider the representations that depended on that fact.

Knowledge systems are generally optimized for accumulation, reuse, and inference. Reality-constrained systems must also be designed for withdrawal. Without that capacity, yesterday's error remains active inside today's coherent output.

Related Phrases and Concepts

- truth maintenance systems
- belief revision
- knowledge-base revision
- nonmonotonic reasoning
- dependency tracking
- correction propagation
- stale inference

- correction debt

Semantic Fidelity Project Resources

- Research Library (GitHub)
- Articles & Essays (Substack)
- Semantic Fidelity Glossary